

The Coefficient-Algebra Spectrum of the Real Affine Group

A full candidate solution to the $ax + b$ conjecture in arXiv:1112.4878

Automated mathematics run `fa_banach_001`, lane 1

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Status. Full solution candidate, likely valid, awaiting human review. The packet proves the concrete conjecture in Spronk–Stokke’s $ax + b$ example: the coefficient-norm algebra generated by the positive irreducible representation has no characters beyond the closed upper-half-plane complexification found for its uniform closure. This proves spectral naturality for this representation; it does not prove the paper’s broader Conjecture 3.9(ii).

1 Source problem

The source is N. Spronk and R. Stokke, *Matrix coefficients of unitary representations and associated compactifications*, arXiv:1112.4878, published in *Indiana University Mathematics Journal* **62** (2013), 99–148 [1]. Its final example studies the real affine group

$$G = \{(a, b) : a > 0, b \in \mathbb{R}\}, \quad (a, b)(a', b') = (aa', ab' + b),$$

and the irreducible representation on $H = L^2(\mathbb{R}_{>0}, ds)$

$$(\pi_+(a, b)f)(s) = a^{1/2}e^{ibs}f(as). \tag{1}$$

Write $A(\pi_+)$ for the closure, in Fourier–Stieltjes norm, of the algebra generated by the matrix coefficients of π_+ . The source defines

$$\tilde{G} = \{(a, z) : a > 0, z \in \mathbb{H}\}, \quad \mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z \geq 0\},$$

with $(a, z)(a', z') = (aa', az' + z)$, and extends (1) to contractions

$$(\tilde{\pi}_+(a, z)f)(s) = a^{1/2}e^{izs}f(as). \tag{2}$$

On page 37, after checking that these contractions are characters, it asks whether they are all the characters.

The exact conjecture is

$$\Phi_{A(\pi_+)} \cong \tilde{G}.$$

The source also proves that the spectrum of the corresponding uniform algebra is $\tilde{G}$. Thus its conjecture says exactly that π_+ is spectrally natural.

We have for f, g, h, k in L^2 that

$$\begin{aligned}
 \langle \pi_+ \otimes \pi_+(a, b) f \otimes g | h \otimes k \rangle &= \langle \pi_+(a, b) f | h \rangle \langle \pi_+(a, b) g | k \rangle \\
 &= a \int_{\mathbb{R}^{>0}} \int_{\mathbb{R}^{>0}} e^{ib(s+t)} f(as) g(at) \overline{h(s)k(t)} dt ds \\
 (4.7) \quad &= a^{1/2} \int_{\mathbb{R}^{>0}} \int_{\mathbb{R}^{>0}} a^{1/2} e^{ibs} f(a(s-t)) g(at) \overline{h(s-t)k(t)} ds dt \\
 &= a^{1/2} \int_{\mathbb{R}^{>0}} \langle \pi_+(a, b) \lambda(at) f g(at) | \lambda(t) h k(t) \rangle dt \\
 &= \langle \Pi_+(a, b) U f \otimes g | U h \otimes k \rangle.
 \end{aligned}$$

In particular Π_+ is a representation, unitarily equivalent to $\pi_+ \otimes \pi_+$. Thus Theorem 3.6 (3.3) tells us that for x in VN_{π_+} we have that $x \in \Phi_{\mathcal{A}(\pi_+)} \cup \{0\}$ if and only if

$$(4.8) \quad U^* \Pi_+^{\pi_+}(x) U = x \otimes x.$$

We remark that since π_+ is irreducible, $\text{VN}_{\pi_+} = \mathcal{B}(L^2)$. We let

$$\tilde{G} = \{(a, z) =: a \in \mathbb{R}^{>0}, z \in \mathbb{H}\}$$

which is a semigroup via $(a, z)(a', z') = (aa', az' + z)$. For (a, z) in $\tilde{G}$, define $\tilde{\pi}_+(a, z)$ exactly as in (4.6). A straightforward repeat of (4.7) shows that each $\tilde{\pi}_+(a, z)$ satisfies (4.8). We conjecture that $\Phi_{\mathcal{A}(\pi_+)} \cong \tilde{G}$. We observe that the conjugation $(a, z)^* = (a^{-1}, -a^{-1}\bar{z})$ satisfies $\tilde{\pi}_+((a, z)^*) = \tilde{\pi}_+(a, z)^*$. We have that $(a, z)^*(a, z) = (1, 2a^{-1}\text{Im}z)$. It is clear that $\tilde{\pi}_+(1, 2a^{-1}\text{Im}z)$ is the operator of multiplication by $x \mapsto e^{-2a^{-1}\text{Im}z x}$ on L^2 , whose positive square root is the operator of multiplication by $x \mapsto e^{-a^{-1}\text{Im}z x}$, i.e. $|\tilde{\pi}_+(a, z)| = \tilde{\pi}_+(1, a^{-1}\text{Im}z)$. Thus we obtain a formula for the polar decomposition from Theorem 3.6 (a, z) = (a, Re z)(1, a^{-1}Im z).

We now claim that

$$(4.9) \quad \mathcal{A}(\pi_+) \cong \mathcal{C}_0(\mathbb{R}^{>0}) \tilde{\otimes} \mathcal{A}_0(\mathbb{H})$$

where $\tilde{\otimes}$ denotes the injective tensor product and $\mathcal{A}_0(\mathbb{H})$ is defined in Example 4.8 (i). Indeed, $\mathcal{A}(\pi_+) \subset \mathcal{C}_0(G) \cong \mathcal{C}_0(\mathbb{R}^{>0} \times \mathbb{R})$ is a point-separating subalgebra which has the span of functions $(a, b) \mapsto a^{1/2} \int_{\mathbb{R}^{>0}} e^{ibs} f(as) g(s) ds$ where $f, g \in L^2$. For fixed b , such a function is easily seen to be a generic element of $\mathcal{A}(\mathbb{R}^{>0})$, the space of which is dense in $\mathcal{C}_0(\mathbb{R}^{>0})$. For a fixed a , such a function may be seen to be the Laplace transform $b \mapsto \hat{h}(b)$, of a generic element h in $L^1(\mathbb{R}^{>0}, m)$, a set of elements which is dense in $\mathcal{A}_0(\mathbb{H})$, as demonstrated in Example 4.8 (i).

We note that it is immediate that $\Phi_{\mathcal{A}(\pi_+)} \cong \tilde{G}$. Conjecture 3.9 if true, would now tell us that $\Phi_{\mathcal{A}(\pi_+)} \cong \tilde{G}$.

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Figure 1: The operator criterion, the complexified semigroup, and the conjecture in the source paper, PDF page 37.

2 Proof intuition

The tensor square of π_+ becomes very transparent in sum-ratio coordinates

$$r = s + t, \quad u = \frac{s}{s + t}.$$

It is just π_+ in the radial variable r , amplified by the identity in the angular variable u . Consequently a character operator x satisfies

$$\Delta(x) = x \otimes x,$$

where Δ is radial amplification. The right side must therefore commute with every multiplier depending only on $s/(s + t)$.

For a positive group-like operator T , this angular commutation is rigid. If T had an off-diagonal block between two separated radial intervals, one could tensor it with a nonzero positive diagonal

block of T on a sufficiently narrow third interval. The resulting block of $T \otimes T$ would connect two disjoint angular ranges, contradicting angular commutation. Hence T is a multiplication operator. Its group-like equation is then the measurable Cauchy equation $f(s+t) = f(s)f(t)$, so $T = M_{e^{-cs}}$, $c \geq 0$.

For a unitary group-like operator V , angular commutation preserves the operator $Q_1(Q_1+Q_2)^{-1}$, where $Q = M_s$. Applying the bounded imaginary powers of the logit function shows that $V^*QV = \alpha Q$. Therefore V is a dilation followed by a unimodular multiplier. The group-like equation again becomes the Cauchy equation and forces that multiplier to be e^{ibs} . Thus $V = \pi_+(a, b)$.

Finally, the polar decomposition of an arbitrary nonzero character is again group-like. The positive and unitary classifications give

$$x = \pi_+(a, b)M_{e^{-cs}} = \tilde{\pi}_+(a, b + iac),$$

which is exactly the claimed closed-upper-half-plane family.

3 The radial coproduct

Let $K = L^2((0, 1), du)$ and define a unitary $W : H \otimes H \rightarrow L^2(\mathbb{R}_{>0} \times (0, 1), dr du) = H \otimes K$ by

$$(W\xi)(r, u) = r^{1/2}\xi(ru, r(1-u)). \quad (3)$$

The Jacobian of $(r, u) \mapsto (ru, r(1-u))$ is r , so W is indeed unitary. A direct calculation gives

$$W(\pi_+ \otimes \pi_+)(a, b)W^* = \pi_+(a, b) \otimes I_K. \quad (4)$$

In particular $\pi_+ \otimes \pi_+$ is an amplification of π_+ , so the closed coefficient space of π_+ is already an algebra and equals $A(\pi_+)$. The normal $*$ -homomorphism occurring in the Walter criterion of Spronk–Stokke is

$$\Delta(x) = W^*(x \otimes I_K)W \quad (x \in \mathcal{B}(H)). \quad (5)$$

That criterion says

$$x \in \Phi_{A(\pi_+)} \cup \{0\} \iff \Delta(x) = x \otimes x. \quad (6)$$

Let

$$R = \frac{Q \otimes I}{Q \otimes I + I \otimes Q}, \quad (Qf)(s) = sf(s).$$

Under W , the operator R is $I_H \otimes M_u$. Thus (5) commutes with every bounded Borel function of R . We record the consequence used repeatedly:

$$\Delta(x) = x \otimes x \implies [x \otimes x, \psi(R)] = 0 \quad (\psi \in L^\infty(0, 1)). \quad (7)$$

Also Δ is isometric. Taking norms in (6) shows that every nonzero group-like operator has norm one.

4 Positive group-like operators

Lemma 1 (Angular block rigidity). *Let $0 \neq T \in \mathcal{B}(H)$ be positive. If $T \otimes T$ commutes with every bounded Borel function of R , then T is a multiplication operator. Consequently, if in addition $\Delta(T) = T \otimes T$, then*

$$T = M_{e^{-cs}}$$

for a unique $c \geq 0$.

Proof. For a Borel set $E \subset \mathbb{R}_{>0}$ write $P_E = M_{1_E}$. First take bounded open intervals $I = (i_0, i_1)$ and $J = (j_0, j_1)$ with $i_1 < j_0$. We prove $P_I T P_J = 0$.

Choose q with $1 < q < j_0/i_1$ and partition $\mathbb{R}_{>0}$, up to endpoints, into the intervals $K_n = (q^n, q^{n+1})$. At least one $K = K_n$ satisfies

$$P_K T P_K \neq 0. \quad (8)$$

Indeed, if every such compression vanished, positivity would give $T^{1/2} P_K = 0$ for every K , hence $T = 0$.

On $I \times K$, the angular coordinate $s/(s+t)$ is at most $i_1/(i_1 + q^n)$. On $J \times K$, it is at least $j_0/(j_0 + q^{n+1})$. The choice $q < j_0/i_1$ makes the first number strictly smaller than the second. Hence there is a Borel set $S \subset (0, 1)$ such that the angular projection $E = 1_S(R)$ is the identity on $L^2(I \times K)$ and zero on $L^2(J \times K)$. Since $T \otimes T$ commutes with E ,

$$\begin{aligned} 0 &= (P_I \otimes P_K) E (T \otimes T) (I - E) (P_J \otimes P_K) \\ &= (P_I T P_J) \otimes (P_K T P_K). \end{aligned}$$

By (8), $P_I T P_J = 0$. Self-adjointness gives the reverse off-diagonal block as well.

Exhausting $(0, t)$ and (t, ∞) by separated bounded intervals and taking strong limits now shows

$$P_{(0,t)} T P_{(t,\infty)} = P_{(t,\infty)} T P_{(0,t)} = 0 \quad (t > 0).$$

Thus T commutes with all spectral projections of Q . The multiplication algebra on $L^2(\mathbb{R}_{>0})$ is maximal abelian, so $T = M_f$ for a bounded measurable $f \geq 0$.

If T is group-like, (3) gives

$$\Delta(M_f) = M_{f(s+t)}, \quad M_f \otimes M_f = M_{f(s)f(t)}.$$

Therefore $f(s+t) = f(s)f(t)$ for almost every (s, t) . The standard measurable version of the exponential Cauchy equation (obtained, if needed, by changing f on a null set) gives either $f = 0$ almost everywhere or $f(s) = e^{\kappa s}$. The first case is excluded. Since every nonzero group-like operator has norm one, $\kappa \leq 0$. Writing $c = -\kappa$ proves the assertion and uniqueness. $\square$

An immediate useful consequence is that the only nonzero group-like projection is I .

5 Unitary group-like operators

Lemma 2 (Logarithmic ratio rigidity). *If $V \in \mathcal{B}(H)$ is unitary and $\Delta(V) = V \otimes V$, then there are unique $a > 0$ and $b \in \mathbb{R}$ such that*

$$V = \pi_+(a, b) = M_{e^{ibs}} D_a, \quad (D_a f)(s) = a^{1/2} f(as).$$

Proof. Put $A = V^* Q V$. From (7), conjugation by $V \otimes V$ fixes R , and hence

$$\frac{A \otimes I}{A \otimes I + I \otimes A} = \frac{Q \otimes I}{Q \otimes I + I \otimes Q}. \quad (9)$$

The function $h_t(r) = (r/(1-r))^{it}$ is bounded on $(0, 1)$ for each $t \in \mathbb{R}$. Applying bounded Borel functional calculus to (9) gives

$$A^{it} \otimes A^{-it} = Q^{it} \otimes Q^{-it}. \quad (10)$$

After multiplying on the left by $Q^{-it} \otimes Q^{it}$, this has the form $C_t \otimes D_t = I$. If two nonzero bounded operators have tensor product equal to the identity, each is scalar; for example, slice by a vector functional nonzero on D_t . Hence

$$Q^{-it} A^{it} = c(t)I, \quad A^{it} = c(t)Q^{it},$$

where $|c(t)| = 1$. Strong continuity and the group laws show that c is a continuous character of $\mathbb{R}$, so $c(t) = e^{i\gamma t}$ for some $\gamma \in \mathbb{R}$. Uniqueness of generators of one-parameter unitary groups then yields

$$A = e^\gamma Q.$$

Set $a = e^{-\gamma}$. Since $D_a^* Q D_a = Q/a = e^\gamma Q$, the unitary $U = V D_a^*$ commutes with Q . Maximal abelianness of the multiplication algebra gives $U = M_\varphi$ for a measurable unimodular function φ . Thus $V = M_\varphi D_a$.

Both V and $D_a = \pi_+(a, 0)$ are group-like. Cancelling $D_a \otimes D_a$ from their group-like equations gives

$$\varphi(s+t) = \varphi(s)\varphi(t) \quad \text{for almost every } (s, t).$$

A measurable circle-valued solution of this Cauchy equation is $\varphi(s) = e^{ibs}$ for a unique $b \in \mathbb{R}$. This proves the result. $\square$

6 Completion of the character classification

Theorem 3 (The $ax + b$ coefficient spectrum). *For the representation (1), every nonzero solution of the Walter equation (6) is uniquely of the form*

$$\tilde{\pi}_+(a, z), \quad a > 0, \quad \text{Im } z \geq 0.$$

Consequently

$$\Phi_{A(\pi_+)} \cong \tilde{G} = \{(a, z) : a > 0, \text{Im } z \geq 0\}$$

as a locally compact topological semigroup. In particular π_+ is spectrally natural.

Proof. Let $0 \neq x$ satisfy (6), and write its polar decomposition as $x = v|x|$. Normal $*$ -homomorphisms preserve absolute values, and

$$|x \otimes x| = |x| \otimes |x|.$$

Therefore $|x|$ is group-like. The polar-decomposition part of the Walter criterion in [1, Theorem 3.6] also gives that v is group-like. Lemma 1 gives

$$|x| = M_{e^{-cs}} \quad (c \geq 0).$$

The initial and final projections v^*v and vv^* are nonzero group-like projections. The consequence following Lemma 1 says both are I , so v is unitary. Lemma 2 now gives $v = \pi_+(a, b)$ for some $a > 0$, $b \in \mathbb{R}$. Hence

$$\begin{aligned} (xf)(s) &= (M_{e^{ibs}} D_a M_{e^{-cs}} f)(s) \\ &= a^{1/2} e^{ibs} e^{-cas} f(as) = \tilde{\pi}_+(a, b + iac) f(s). \end{aligned}$$

Since $b + iac \in \mathbb{H}$, every nonzero group-like operator belongs to the stated family. Conversely, substituting (2) in (3) directly gives

$$\Delta(\tilde{\pi}_+(a, z)) = \tilde{\pi}_+(a, z) \otimes \tilde{\pi}_+(a, z),$$

as already observed in the source. Uniqueness follows first from support transport, which determines a , and then from equality of the exponential multipliers, which determines z .

It remains only to record the topology implicit in $\cong$. The map

$$(a, z) \mapsto \tilde{\pi}_+(a, z) \tag{11}$$

is weak-operator continuous. Adjoin one point ∞ to $\tilde{G}$ and send it to the zero operator. This extension is continuous: for compactly supported test vectors bounded away from zero, coefficients vanish identically when $a \rightarrow 0$ or $a \rightarrow \infty$, vanish by domination when $\text{Im } z \rightarrow \infty$, and vanish by the Riemann–Lebesgue lemma when $|\text{Re } z| \rightarrow \infty$ while the other parameters stay in compact sets. Density and contractivity remove the support restriction.

The set of solutions to (6) in the operator unit ball is weak-operator closed: Δ is weak-operator continuous on bounded sets, and $x \mapsto x \otimes x$ is weak-operator continuous there on elementary matrix coefficients. It is therefore compact. The extended map from the one-point compactification of $\tilde{G}$ is, by the classification above, a continuous bijection onto this compact solution set. It is a homeomorphism. Removing the point mapped to zero proves the topological assertion. The semigroup law follows directly from

$$\tilde{\pi}_+(a, z)\tilde{\pi}_+(a', z') = \tilde{\pi}_+(aa', az' + z).$$

Finally, the source proves $\Phi_{\mathcal{A}(\pi_+)} \cong \tilde{G}$ for the uniform closure $\mathcal{A}(\pi_+)$. Thus equality of the two spectra is exactly spectral naturality of π_+ . $\square$

7 Verification, limitations, and novelty

Internal verification. The proof was checked against the source’s Walter equation, its polar decomposition statement, and the explicit formula for π_+ . The two places where hidden regularity could enter were isolated. The positive classification uses only interval projections, positivity, and strong limits; it never assumes an integral kernel for T . The unitary classification uses only bounded imaginary powers of the ratio operator; no subtraction of unbounded logarithms is performed. The resulting parameter formula agrees with the polar formula displayed by Spronk–Stokke: $z = b + iac$ means $c = a^{-1} \text{Im } z$.

Scope limitation. This settles the explicit conjecture in the $ax + b$ example and hence one important case of the paper’s Conjecture 3.9(ii), because $\pi_+ \leq_q \lambda_G$. It does not prove that every representation quasi-contained in a left regular representation is spectrally natural.

Bounded novelty search. On 26 August 2026, the four run indexes and the local arXiv source corpus were searched for arXiv:1112.4878, “spectrally natural”, “Wiener–Pitt representation”, and the $\pi_+/ax + b$ character-space conjecture. Web and arXiv searches used the exact title, exact conjecture phrase, the terms “spectrally natural”, “ $ax + b$ coefficient algebra spectrum”, and close variants. They found the source article, its journal record, citations that use its general compactification framework, and no later paper stating a resolution of this conjecture. This is a bounded rather than exhaustive novelty check.

Human review recommendation. Send to a harmonic analyst or operator algebraist. The highest-value checks are Lemma 1’s passage from separated interval blocks to the multiplication masa, and Lemma 2’s imaginary-power functional calculus. Neither step currently has an identified gap.

References

- [1] N. Spronk and R. Stokke, *Matrix coefficients of unitary representations and associated compactifications*, Indiana Univ. Math. J. **62** (2013), no. 1, 99–148; arXiv:1112.4878.